MORE ON DATAFRAMES

A notebook for an exploratory analysis of TED talks data at:

https://colab.research.google.com/drive/1Ohqj99tdIPNkZ6fsvrKs6jjVMJ335S8X?usp=sharing

DATA WRANGLING

Reading: Ch3 from Skiena

- Acquiring data and preparing it for analysis
- Data Pipelines:

  Notebooks make it easier to maintain data pipelines, the sequence of processing steps from start to finish.

  Expect to have to redo your analysis from scratch, so build your code to make it possible.

LANGUAGES

- Python: many libraries and features (e.g regular expressions)
- R: programming language of statisticians.
- Matlab: fast and efficient matrix operations.
- Java/C: for Big Data systems.
- Excel: good for EDA
- …

COMMON DATA FORMATS

- CSV (comma separated value):

A small part of ted_main.csv file we have seen earlier


```
comments,description,duration,event,film_date,languages,main_speaker,name,num_speaker,published_date,ratings,related_talks,speaker_occupation,tags,title,url,views
4553,Sir Ken Robinson makes an entertaining and profoundly moving case for creating an education system that nurtures (rather than undermines)
creativity.,1164,TED2006,1140825600,60,Ken Robinson,Ken Robinson: Do schools kill creativity?,1,1151367060,"[{'id': 7, 'name': 'Funny', 'count': 19645}, {'id': 1, 'name':
'Beautiful', 'count': 4573}, {'id': 9, 'name': 'Ingenious', 'count': 6073}, {'id': 3, 'name': 'Courageous', 'count': 3253}, {'id': 11, 'name': 'Longwinded', 'count': 387},
{'id': 2, 'name': 'Confusing', 'count': 242}, {'id': 8, 'name': 'Informative', 'count': 7346}, {'id': 22, 'name': 'Fascinating', 'count': 10581}, {'id': 21, 'name':
'Unconvincing', 'count': 300}, {'id': 24, 'name': 'Persuasive', 'count': 10704}, {'id': 23, 'name': 'Jaw-dropping', 'count': 4439}, {'id': 25, 'name': 'OK', 'count': 1174},
{'id': 26, 'name': 'Obnoxious', 'count': 209}, {'id': 10, 'name': 'Inspiring', 'count': 24924}]","[{'id': 865, 'hero': 'https://pe.tedcdn.com/images/ted/172559_800x600.jpg',
'speaker': 'Ken Robinson', 'title': 'Bring on the learning revolution!', 'duration': 1008, 'slug': 'sir_ken_robinson_bring_on_the_revolution', 'viewed_count': 7266103},
{'id': 1738, 'hero': 'https://pe.tedcdn.com/images/ted/de98b161ad1434910ff4b56c89de71af04b8b873_1600x1200.jpg', 'speaker': 'Ken Robinson', 'title': ""How to escape
education's death valley"", 'duration': 1151, 'slug': 'ken_robinson_how_to_escape_education_s_death_valley', 'viewed_count': 6657572}, {'id': 2276, 'hero': 'https://
pe.tedcdn.com/images/ted/3821f3728e0b755c7b9aea2e69cc093eca41abe1_2880x1620.jpg', 'speaker': 'Linda Cliatt-Wayman', 'title': 'How to fix a broken school? Lead fearlessly,
love hard', 'duration': 1027, 'slug': 'linda_cliatt_wayman_how_to_fix_a_broken_school_lead_fearlessly_love_hard', 'viewed_count': 1617101}, {'id': 892, 'hero': 'https://
pe.tedcdn.com/images/ted/e79958940573cc610ccb583619a54866c41ef303_2880x1620.jpg', 'speaker': 'Charles Leadbeater', 'title': 'Education innovation in the slums', 'duration':
1138, 'slug': 'charles_leadbeater_on_education', 'viewed_count': 772296}, {'id': 1232, 'hero': 'https://pe.tedcdn.com/images/ted/
```

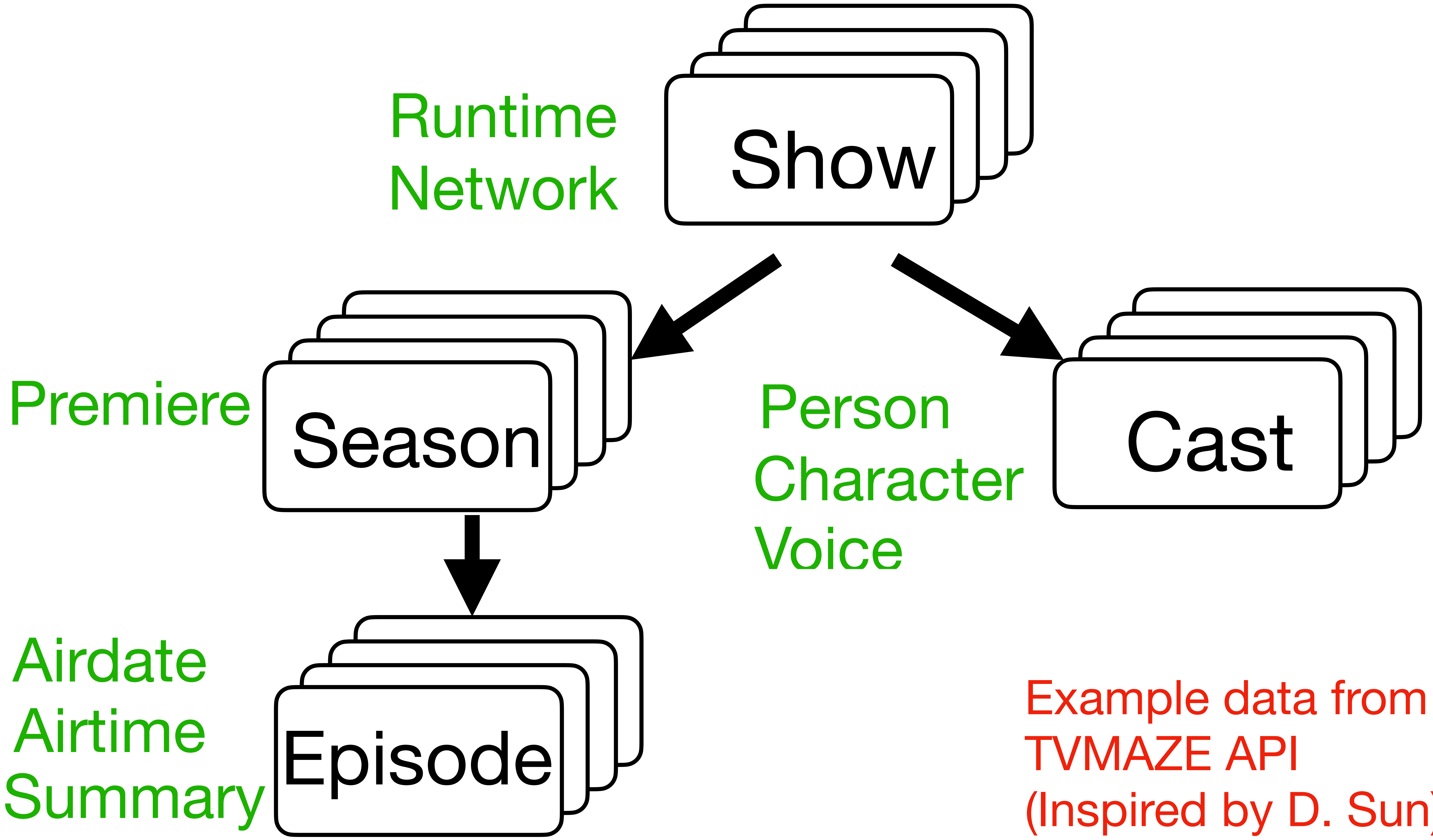

Can easily convert to a pandas dataframe for further processing:

```
ted_main = pd.read_csv('/content/drive/My Drive/datasets/ted_main.csv')
```

COMMON DATA FORMATS

- JSON (JavaScript Object Notation):

  [{'name': ‘Black Mirror’, ‘runtime': 60, 'network': {'name': '', ...}
  'cast': [{'person': {'name': ...}, 'character': {'name': ...}, 'voice': False}, …],
  'seasons': [{'premiereDate': '2012-04-15', 'episodes': [...]}, ...]}, …]



Example data from TVMAZE API (Inspired by D. Sun)

Hierarchical data

COMMON DATA FORMATS

- XML (eXtensible Markup Language):

- Like JSON, can represent hierarchical data:
    - Fields shown with tags
    - Each tag with open <…> and close </…>
    - Children shown with nested tags
    - Repeated fields shown with repeated tags

- Can use BeautifulSoup library to read XML data into a BeautifulSoup object, which represents the data as a tree.

```
<?xml version="1.0" encoding="UTF-8"?>
<root>
  <show>
    <name>Black Mirror</name>
    <runtime>60</runtime>
    <cast>
      <person>...</person>
      <character>....</character>
    </cast>
    <cast>

       ...

    </cast>
    <season>
      <episode>...</episode>
      <episode>...</episode>
      ...
    </season>
    <season>
       ...
    </season>
  </show>
</root>
```

Example data
Inspired by D. Sun

DATA SOURCES

- Academic data sets
  Some journals require datasets to be published, some digging will be necessary. Economic, medical, demographic, historical, and scientific data.

- Sensor data

- Other important data sources in more detail in the next slides…

DATA SOURCES

- Government data sets
  Data from City, State, and Federal governments. May differ from city/state…

ST_NUM	CITY	ZIP_CODE	OWN_OCC	OWNER	MAIL_STREET_ADDRESS	LAND_SF	GROSS_AREA	LIVING_AREA	LAND_VALUE	BLDG_VALUE	TOTAL_VALUE	GROSS_TAX	YR_BUILT	YR_REMODEL
104	EAST BOSTON	02128	Y	PASCUCCI CARLO	195 LEXINGTON ST	1,150	3353	2202	197,600	594,400	792,000	$8,632.80	1900	
197	EAST BOSTON	02128	N	SEMBRANO RODERICK	197 LEXINGTON ST	1,150	3047	2307	198,500	619,700	818,200	$8,918.38	1920	2000
199	EAST BOSTON	02128	Y	GUERRA CHEVARRIA ANA S	199 LEXINGTON ST	1,150	3392	2268	199,100	605,300	804,400	$8,767.96	1905	1985
201	EAST BOSTON	02128	N	JB REALTY TRUST	PO BOX 557 #	1,150	3108	2028	199,700	535,600	735,300	$8,014.77	1900	1991
203	EAST BOSTON	02128	Y	MARKS TRAVIS JOSEPH	203 Lexington ST	2,010	3700	2546	230,200	501,400	731,600	$7,974.44	1900	1978
205	EAST BOSTON	02128	N	205 LEXINGTON LLC	28 LAUDHOLM RD	2,500	6278	4362	263,800	1,037,400	1,301,200	$14,183.08	1900	2018
209	EAST BOSTON	02128	N	YOON SUNG PIL	-211 209 LEXINGTON ST	2,500	6432	4296	264,700	1,003,200	1,267,900	$13,820.11	1900	2009
213	EAST BOSTON	02128	Y	CASTALDINI ANTONIO	213 LEXINGTON ST	2,500	6048	4080	265,300	885,400	1,150,700	$12,542.63	1900	

PROPERTY ASSESSMENT data from data.boston.gov. For instance, no such data available for Tucson?. Neighborhood income doesn't have this information at this detail level.

- data.gov has almost 300K datasets available!

DATA SOURCES

- Proprietary data sources
  Usually via rate-limited application program interfaces (APIs)
- For an example check:

https://colab.research.google.com/drive/19pC9y6DpN-Y5677kr6AuJ18Jff2QOgkl?usp=sharing

```python
import requests
jmirror = requests.get("http://api.tvmaze.com/search/shows?q=mirror")
jmirror.text[:500]
```


'[{"score":0.7032747,"show":{"id":305,"url":"https://www.tvmaze.com/sh
ish","genres":["Drama","Science-Fiction","Thriller"],"status":"Running
ficialSite":"https://www.netflix.com/title/70264888","schedule":{"time
el":{"id":1,"name":"Netflix","country":null,"officialSite'


```python
pmirror = jmirror.json()
pmirror
```


[{'score': 0.7032747,
  'show': {'id': 305,
   'url': 'https://www.tvmaze.com/shows/305/black-mirror',
   'name': 'Black Mirror',
   'type': 'Scripted',
   'language': 'English',
   'genres': ['Drama', 'Science-Fiction', 'Thriller'],
   'status': 'Running',
   'runtime': None,
   'averageRuntime': 63,


```python
import pandas as pd
df_mirror = pd.json_normalize(pmirror)
df_mirror.head()
```

DATA SOURCES

- Scraping websites: Which US town has seen the largest growth 2010-2020?
- For an example check:

  https://colab.research.google.com/drive/1pcsmyP0fcq8kL14Y4hy47StocV4jvF_f?usp=sharing

```python
import requests
response = requests.get(
    "https://en.wikipedia.org/wiki/Tucson,_Arizona")
```

```python
from bs4 import BeautifulSoup
soup = BeautifulSoup(response.text, "html.parser")
tucson_hist_pop = soup.find('table',{'class':'us-census-pop us-census-pop-right'})
```

```python
tucson_hist_pop
```

```
<table class="us-census-pop us-census-pop-right">
<caption>Historical population</caption>
<tbody><tr><th scope="col">Census</th><th scope="col"><abbr title="Population">Pop.</abb
deduplicate="TemplateStyles:r1152813436">.mw-parser-output .sr-only{border:0;clip:rect(0
0px);height:1px;margin:-1px;overflow:hidden;padding:0;position:absolute;width:1px;white-
<th scope="col"><abbr title="Percent change">%±</abbr></th></tr>
<tr><th scope="row"><a href="/wiki/1850_United_States_census" title="1850 United States
<tr><th scope="row"><a href="/wiki/1860_United_States_census" title="1860 United States
```

```python
data = []
for row in tucson_hist_pop.find_all('tr'):
    row_data = []
    for cell in row.find_all('td'):
        row_data.append(cell.text)
    data.append(row_data)
```

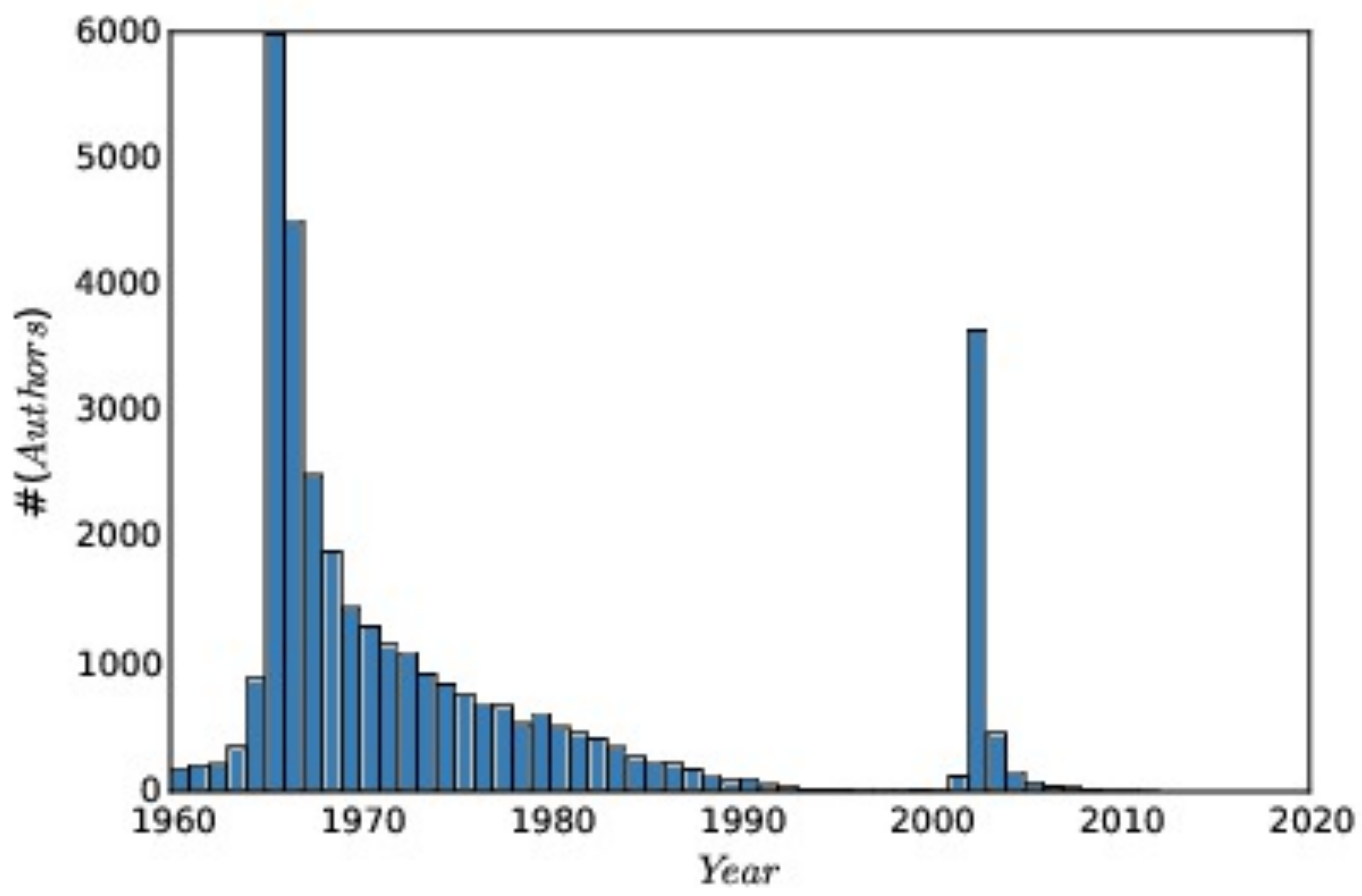
DATA CLEANING

- Many potential issues:
  - Distinguishing errors from artifacts.
  - Data compatibility / unification.
  - Imputation of missing values.
  - Estimating unobserved (zero) counts.
  - Outlier detection.

Image from ExcelKid

DATA CLEANING

- Errors: Information lost in acquisition (measurements etc.)
- Artifacts: Problems arising from processing done to raw data.



100,000 most prolific authors, binned according to the year of their first paper appearing in Pubmed

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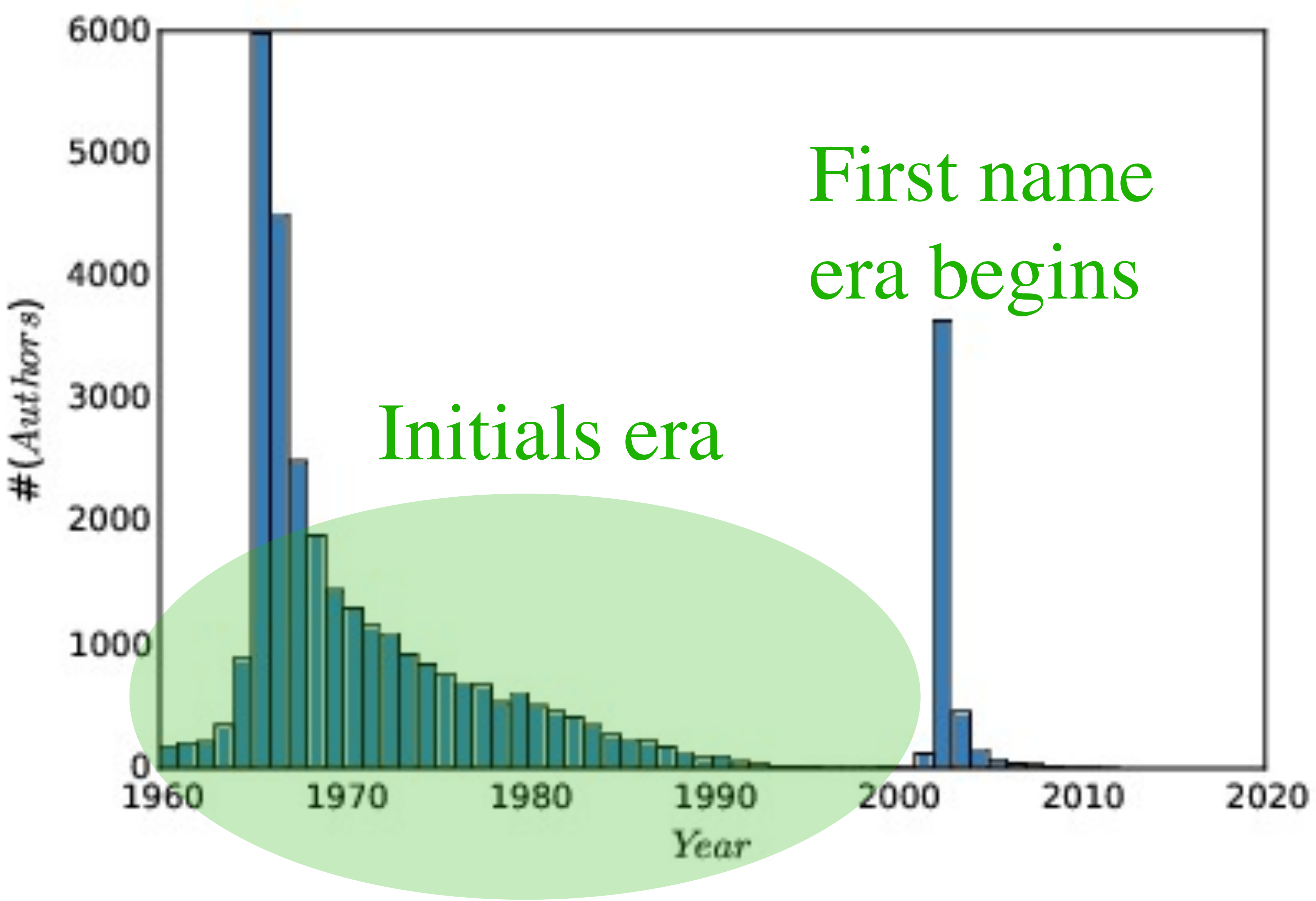

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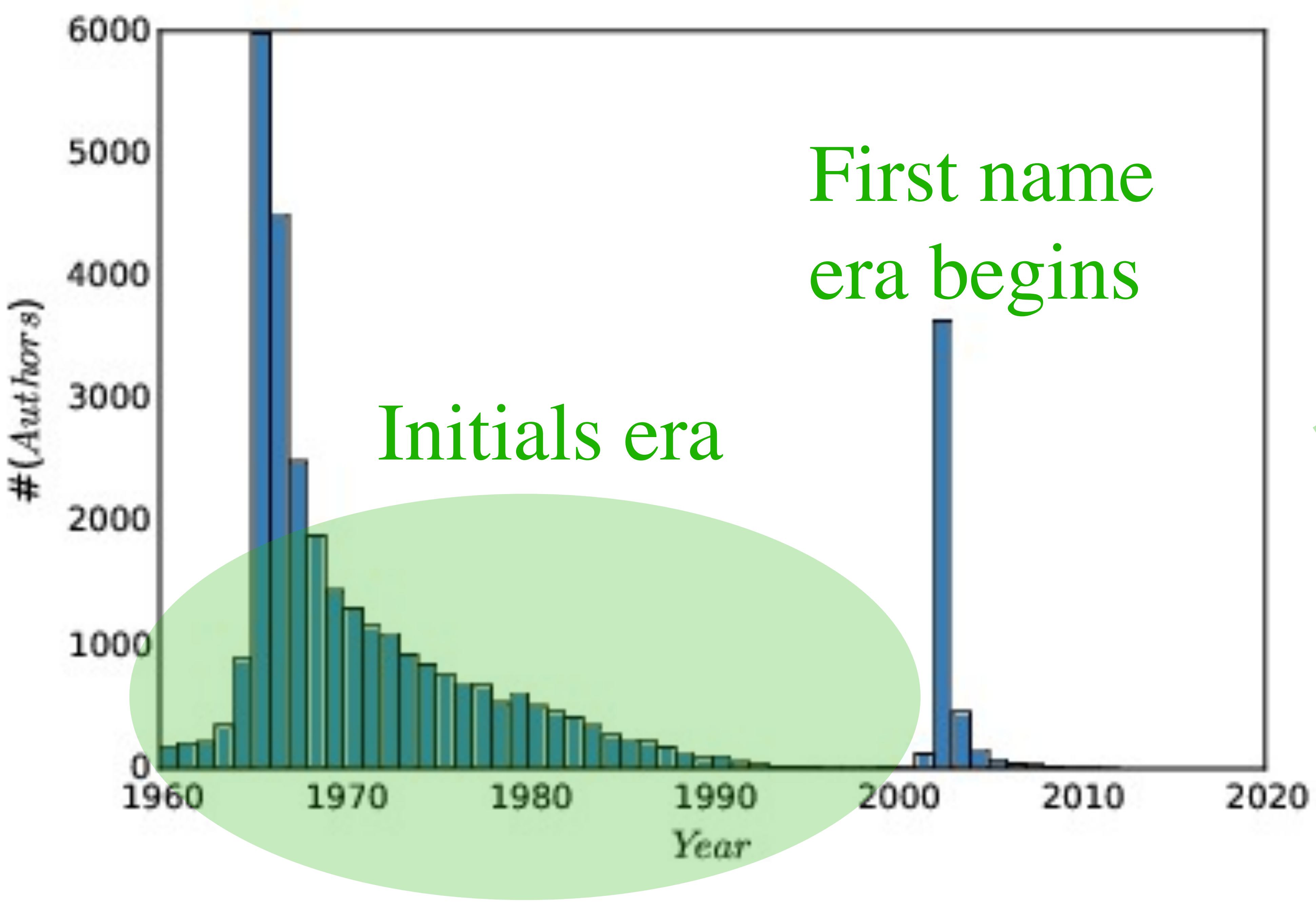

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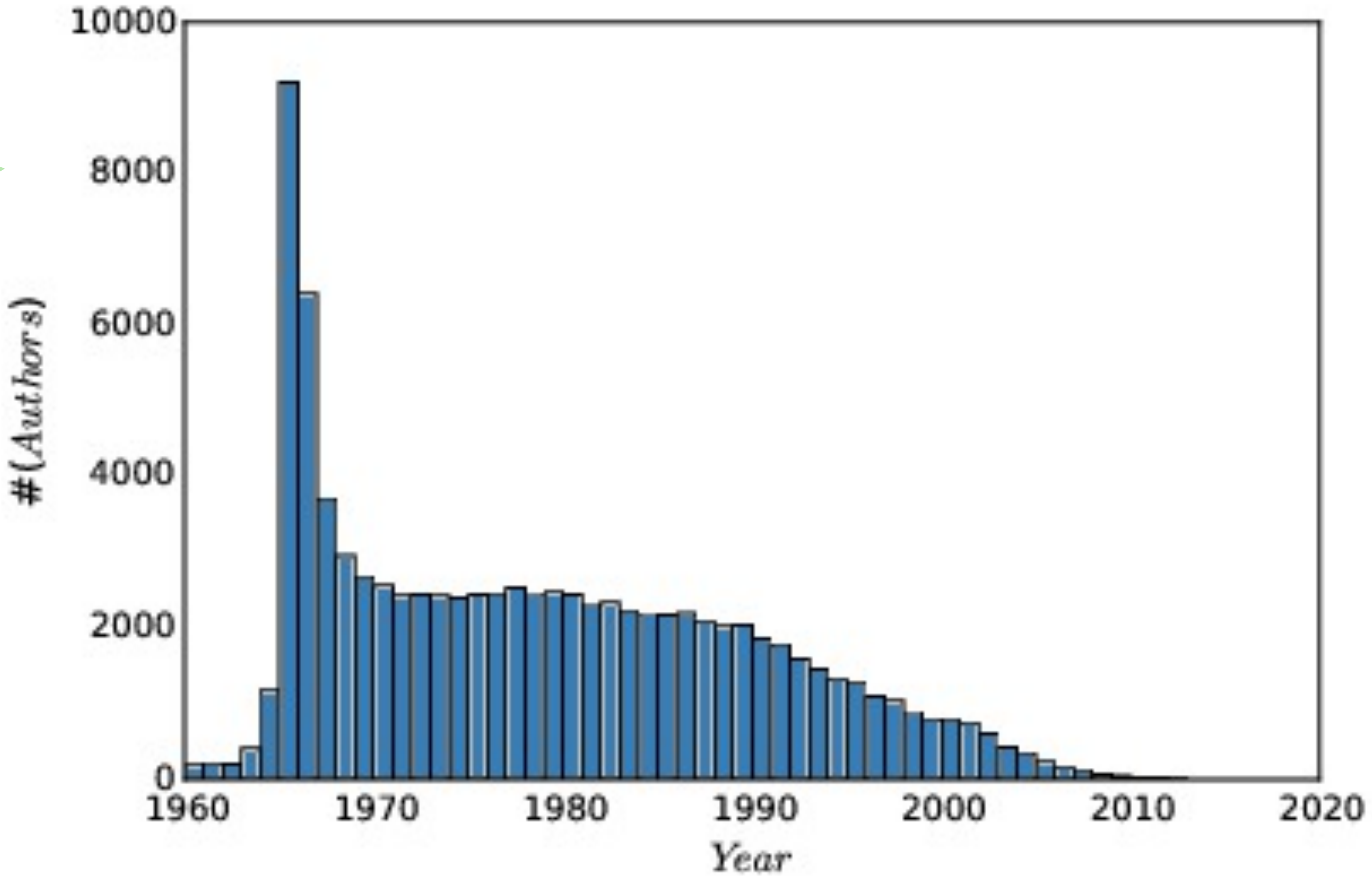


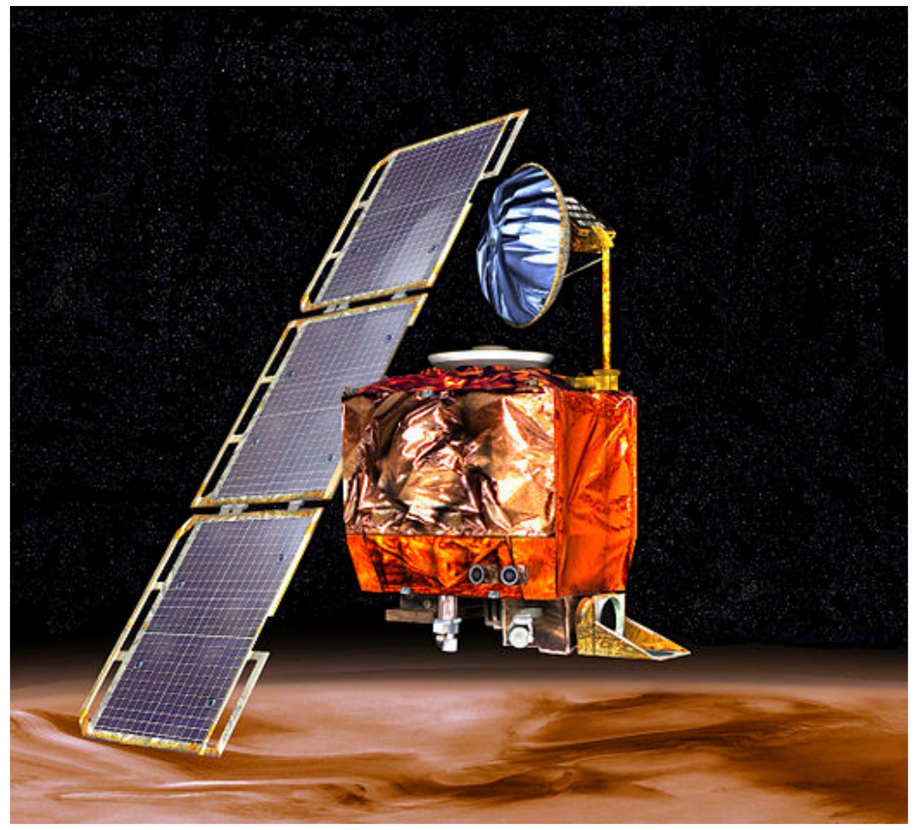
After cleaning the data (merging different strings representing same names)

DATA CLEANING

- Data compatibility issues to be aware of:
  - Unit conversions
  - Number / character code representations
  - Name unification
  - Time/date unification
  - Financial unification

DATA CLEANING

- Unit conversions
  - Can have disastrous effects: Mars Climate Orbiter

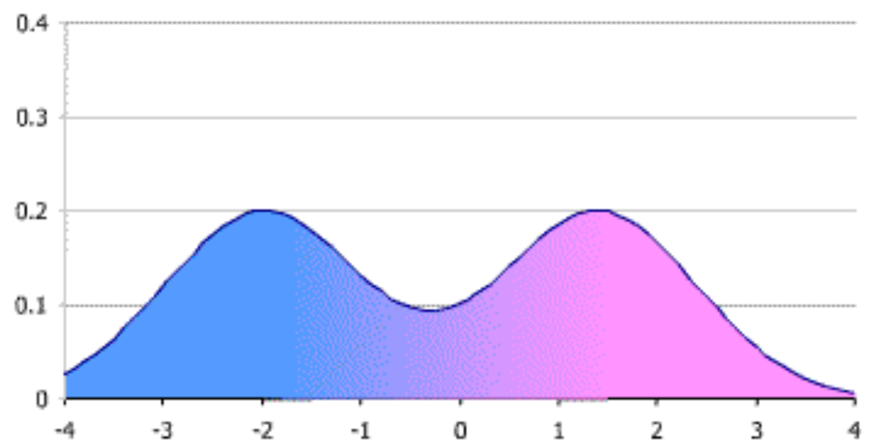


entered an orbit around the Sun.[2] An investigation attributed the failure to a measurement mismatch between two measurement systems: SI units (metric) by NASA and US customary units by spacecraft builder Lockheed Martin.[3]

from Wikipedia

Even if decide on metric: cm, m, km?

  - For same variable: watch out if there is a bimodal distribution



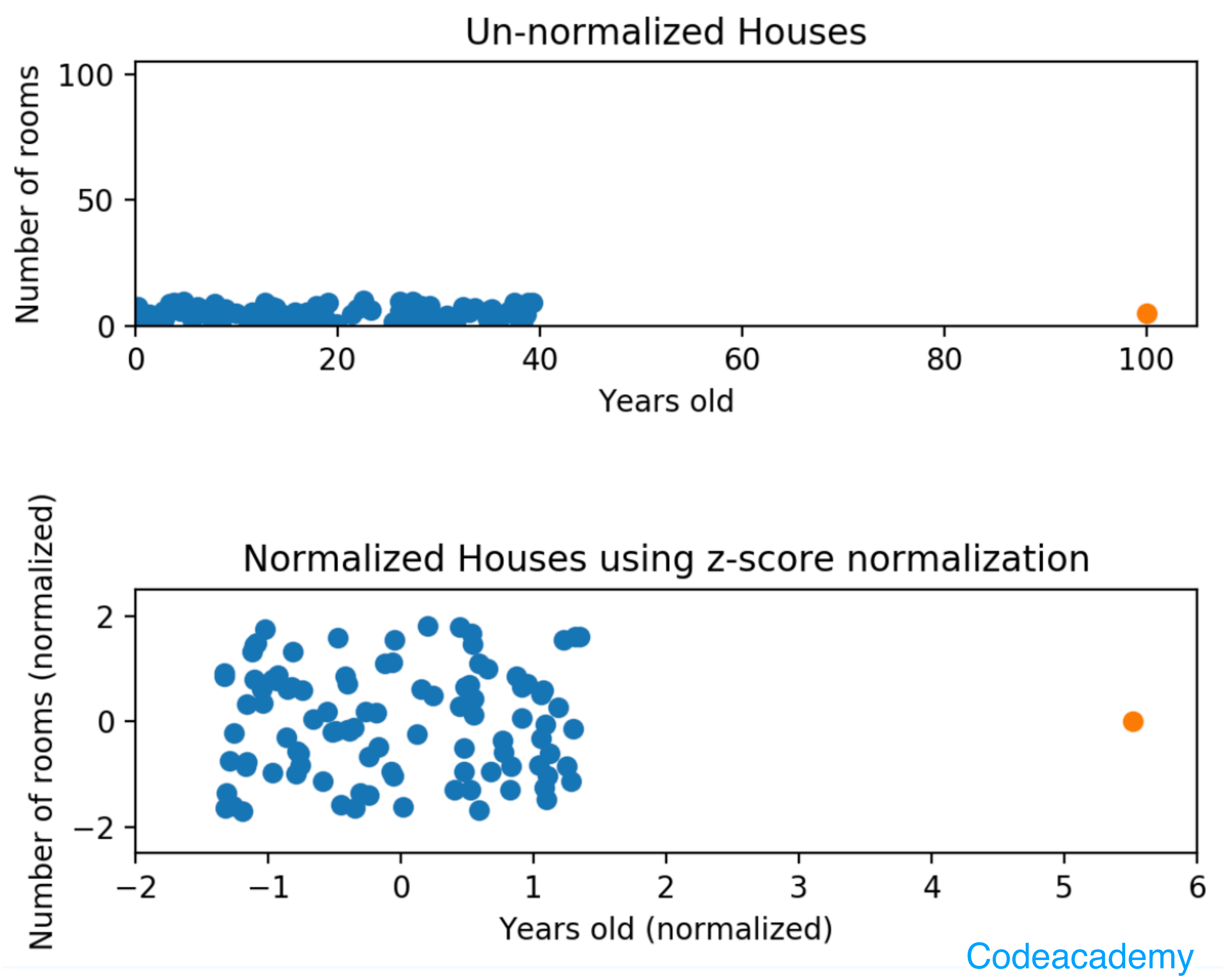
- Multiple variables: Normalization is important.

DATA CLEANING

- Normalization with Z-scores
  - Z-scores are dimensionless quantities
  - Normalize different variables:

    range/distribution comparable.

$$Z_i = (X_i - \overline{X}) / \hat{\sigma}$$



- Z-scores of height in inches same as height measured in miles.
- Z-scores have mean 0 and sigma=1:

$$\mu(B) = 21.9 \quad \sigma(B) = 1.92$$
$$\mu(Z) = 0 \quad \sigma(Z) = 1$$

B	19	22	24	20	23	19	21	24	24	23
Z	-1.51	0.05	1.09	-0.98	0.57	-1.51	-0.46	1.09	1.09	0.57